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ACADEMIC REFLECTIVE JOURNAL

L10 UNSUPERVISED LEARNING

1. INTRODUCTION – (JENNIFER)

In lab 10 Unsupervised learning, there is so much to go over, and I pretty much explored the differences between supervised and unsupervised learning and the ways they can help show how a pattern is processed in data. While using the K-means clustering tool within the group in the four measurements towards the 2 components that help give the results to connect the data with the proper graphs.

2. DESCRIPTION OF EXPIRIENCE/TOPIC - (JENNIFER)

With the supervised learning we know the targets and the trained models, while the unsupervised learning we must do more of the discovering with the K-means models for this lab. Seeing the group points and the clusters with their centers, directions in the PCA being found and capture in variations of directions and even the measurements. Being able to plot the inertia points measuring the sum of a squared distances between the points and their centers to also get the elbow to make the four clusters and just plot them with their centroids.

3. PERSONAL REFLECTION – (MARCUS)

I was a little scared to start this lab with unsupervised learning. It felt like navigating without a map, as we didn't have any labeled targets to work with.

It was very useful to have a private sandbox environment, where I can safely play around with the code and really understand how K-Means and PCA work under the hood.

It was great to see how Elbow Method could find optimal clusters and how PCA could compress high dimensional data and keep the important patterns.

I think overall the opportunity to test concepts on our own helped bridge the gap between raw Python execution and real data science insights.

4. DISCUSSION OF IMPROVEMENT AND LEARNING - (JENNIFER)

Truly the errors I would be getting help me learn on how to resolve and understand them properly and how to approach the problem. But being able to develop new skills such as the K-means models and tools, being able to compare inertia values, and identifying the centroids. Learning all this I could see how to use the PCA to help visualize the datasets given and understanding the meaning and being able to scale what is happening in all the features before and after it's in a group/cluster.

5. CONCLUSION - (JENNIFER)

Overall, lab 10 help me understand how clustering and the reduction in the different sections about the data. While using the K-means revealing the four groups of the clustering data, while also explaining all the process with the coding aspect and the different styles of supervised and unsupervised learning.